

# Z Binding Calculus

txt2tex example

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$[TrackId, AlbumId, ArtistId]$

*Track*

$trackId : TrackId$   
 $albumId : AlbumId$   
 $duration : \mathbb{N}$

*Album*

$albumId : AlbumId$   
 $artistId : ArtistId$   
 $year : \mathbb{N}$   
 $tracks : \mathbb{N}$

## Atomic Binding

$\langle\!\langle trackId == t.trackId \rangle\!\rangle$

## Binding in Set Comprehension

$\{ t : Track \mid t.duration < 180 \bullet \langle\!\langle trackId == t.trackId \rangle\!\rangle \}$

## Multi-Component Binding

$\langle\!\langle trackId == t.trackId, year == a.year, tracks == a.tracks \rangle\!\rangle$

## Multi-Variable Comprehension with Binding

$\{ t : Track; a : Album \mid t.albumId = a.albumId \bullet \langle\!\langle trackId == t.trackId, year == a.year, tracks == a.tracks \rangle\!\rangle \}$

## Empty Binding

$\langle\!\langle \rangle\!\rangle$